

Stacking Spectra of Galaxies and Quasi-Stellar Objects: Enhancing Signal-to-Noise Ratio and Analyzing Active Galactic Nuclei

[Complete code available on GitHub](#)

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Date: 4th September 2024

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Acknowledgements

I would like to express my deepest gratitude to my mentor, Prof. Shadab Alam, for his invaluable guidance, support, and encouragement throughout this summer project. His insights and expertise have greatly enriched my understanding of QSO's as the tracers of our Universe. I am also grateful to TIFR Mumbai for providing the resources and a stimulating environment to pursue this research. Special thanks to my colleagues and peers for their collaboration and constructive feedback. Finally, I extend my heartfelt appreciation to my family and friends for their unwavering support and motivation during this journey.

This research uses services or data provided by the SPectra Analysis and Retrievable Catalog Lab (SPARCL), which is part of the Community Science and Data Center (CSDC) program at NSF's National Optical-Infrared Astronomy Research Laboratory. NOIRLab is operated by the Association of Universities for Research in Astronomy (AURA), Inc. under a cooperative agreement with the National Science Foundation.

Funding for the Sloan Digital Sky Survey IV has been provided by the Alfred P. Sloan Foundation, the U.S. Department of Energy Office of Science, and the Participating Institutions. SDSS-IV acknowledges support and resources from the Center for High Performance Computing at the University of Utah. The SDSS website is www.sdss4.org. SDSS-IV is managed by the Astrophysical Research Consortium for the Participating Institutions of the SDSS Collaboration including the Brazilian Participation Group, the Carnegie Institution for Science, Carnegie Mellon University, Center for Astrophysics — Harvard & Smithsonian, the Chilean Participation Group, the French Participation Group, Instituto de Astrofísica de Canarias, The Johns Hopkins University, Kavli Institute for the Physics and Mathematics of the Universe (IPMU) / University of Tokyo, the Korean Participation Group, Lawrence Berkeley National Laboratory, Leibniz Institut für Astrophysik Potsdam (AIP), Max-Planck-Institut für Astronomie (MPIA Heidelberg), Max-Planck-Institut für Astrophysik (MPA Garching), Max-Planck-Institut für Extraterrestrische Physik (MPE), National Astronomical Observatories of China, New Mexico State University, New York University, University of Notre Dame, Observatório Nacional / MCTI, The Ohio State University, Pennsylvania State University, Shanghai Astronomical Observatory, United Kingdom Participation Group, Universidad Nacional Autónoma de México, University of Arizona, University of Colorado Boulder, University of Oxford, University of Portsmouth, University of Utah, University of Virginia, University of Washington, University of Wisconsin, Vanderbilt University, and Yale University.

This research used data obtained with the Dark Energy Spectroscopic Instrument (DESI). DESI construction and operations is managed by the Lawrence Berkeley National Laboratory. This material is based upon work supported by the U.S. Department of Energy, Office of Science, Office of High-Energy Physics, under Contract No. DE-AC02-05CH11231, and by the National Energy Research Scientific Computing Center, a DOE Office of Science User Facility under the same contract. Additional support for DESI was provided by the U.S. National Science Foundation (NSF), Division of Astronomical Sciences under Contract No. AST-0950945 to the NSF’s National Optical-Infrared Astronomy Research Laboratory; the Science and Technology Facilities Council of the United Kingdom; the Gordon and Betty Moore Foundation; the Heising-Simons Foundation; the French Alternative Energies and Atomic Energy Commission (CEA); the National Council of Science and Technology of Mexico (CONACYT); the Ministry of Science and Innovation of Spain (MICINN), and by the DESI Member Institutions: www.desi.lbl.gov/collaborating-institutions. The DESI collaboration is honored to be permitted to conduct scientific research on Iolkam Du’ag (Kitt Peak), a mountain with particular significance to the Tohono O’odham Nation. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the U.S. National Science Foundation, the U.S. Department of Energy, or any of the listed funding agencies.

1. Summary

This project involves developing a code for stacking spectra of galaxies and QSOs, utilizing data from three major astronomical surveys: SDSS Data Release 16, BOSS Data Release 16, and DESI Early Data Release. The primary objective is to improve the signal-to-noise ratio (SNR) of spectra from galaxies situated in similar environments, particularly based on their position within the dark matter halos of galactic clusters. This enhancement allows for a more detailed study of the properties of their Active Galactic Nuclei (AGNs). Stacking spectra is crucial for minimizing anomalies and facilitating the analysis of emission lines, which in turn enables the estimation of AGN size, star formation rates, chemical composition, and the likelihood of galaxy mergers. Additionally, the stacked spectra provide a reference for identifying unknown galaxies by comparing their spectra to established profiles. This study aims to check whether the pre-established methods in used in the paper Jed McKinney et al (2023)^[1] are able to work for the data given by the three surveys in the optical range.

2. Introduction

The study of galaxies and QSOs through spectral analysis is crucial for understanding the formation and evolution of cosmic structures. This project focuses on stacking spectra from galaxies and QSOs using SPARCL by NOIRLab’s Astro Data Lab.

2.1 SPARCL Client

SPECTRA Analysis & RETRIEVABLE Catalog Lab (SPARCL) at NOIRLab’s Astro Data Lab provides flexible access to spectra from large optical and near-infrared surveys. Major elements of SPARCL include capabilities to discover and query for spectra based on parameters of interest, a fast web service that delivers desired spectra either individually or in bulk. SPARCL currently contains one-dimensional spectroscopic data from the Sloan Digital Sky Survey (SDSS), from both the original SDSS spectrograph and the upgraded instrument of the Baryon Oscillation Spectroscopic Survey (BOSS). SPARCL has also been designed to support spectra from the Dark Energy Spectroscopic Instrument (DESI). The project took the basic format for the code from sparclclient GitHub^[2] page.

2.2 Data Releases

This project is using data from three major astronomical surveys: the Sloan Digital Sky Survey (SDSS) Data Release 16, the Baryon Oscillation Spectroscopic Survey (BOSS) Data Release 16, and the Dark Energy Spectroscopic Instrument (DESI) Early Data Release.

2.2.1 SDSS Data Release 16^[3]

The Sloan Digital Sky Survey (SDSS) is one of the most comprehensive astronomical surveys ever conducted. Its sixteenth data release (DR16) is part of the fourth phase of SDSS, providing spectra for millions of celestial objects, including stars, galaxies, and QSOs. The

spectra in DR16 are well-calibrated, with coverage in the optical region (3800–9200 Å). This data set is ideal for studying the properties of galaxies, particularly their AGNs, due to its extensive redshift coverage and high signal-to-noise ratio.

2.2.2 BOSS Data Release 16

The Baryon Oscillation Spectroscopic Survey (BOSS) is a sub-survey within SDSS, specifically designed to map the large-scale structure of the universe. BOSS DR16 includes spectra from over 1.5 million luminous galaxies and 160,000 QSOs, covering a redshift range up to $z \sim 2.5$. This data is particularly valuable for studying the distribution of dark matter and the evolution of galaxies over cosmic time. BOSS’s spectra, with their high resolution and extensive redshift range, are critical for analyzing galaxies in various environments, especially within the dark matter halos of clusters.

2.2.3 DESI Early Data Release^[4]

The Dark Energy Spectroscopic Instrument (DESI) aims to map the expansion history of the universe by collecting spectra from millions of galaxies and QSOs. The Early Data Release (EDR) of DESI includes data from its first year of operation, offering high-resolution spectra across a wide wavelength range (3600–9800 Å). DESI’s advanced instrumentation enables precise redshift measurements, crucial for studying galaxy clustering and the cosmic web. The high-quality data from DESI EDR is invaluable for stacking spectra and enhancing the SNR in the analysis of galactic and extragalactic objects.

3. Project Goals

The primary goal of this project is to develop a robust, flexible code for stacking spectra of galaxies and QSOs from these three surveys. By improving the signal-to-noise ratio (SNR), the code aims to facilitate the study of AGNs and other key properties of galaxies in similar environments. The code also provides tools for selecting galaxies based on specific criteria, such as redshift, right ascension, declination, and specids (unique identifiers).

4. Methodology

The methodology involves the development of several functions, each with specific roles in the stacking process. Below is a detailed description of each function, including its arguments and return values.

4.1 Data Acquisition (rec function)

This function is responsible for extracting and organizing spectral data for the required galaxies and QSOs from the specified data releases. It changes the wavelength to its loga-

rithmic values and finds the bin size to create bins of the highest resolution (usually of the order $\ln(10) \times 10^{-4}$). This change is made to ensure uniform distribution of data which is only possible on *log* scale

- **Arguments:**

- **result:** The output of the SPARCL query, containing spectral data.
- **limit:** The number of galaxies or QSOs for which data is to be retrieved.

- **Process:**

- **Data Extraction:** The function retrieves the wavelength, flux, and inverse variance arrays for the specified objects using the `rec.wavelength`, `rec.flux`, and `rec.ivar` functions within SPARCL.
- **Redshift Adjustment:** The function adjusts the wavelength values to account for the redshift of each galaxy.
- **Logarithmic Transformation:** Wavelengths are converted to a logarithmic scale for uniform distribution, aligning with the spacing used in the surveys.

- **Returns:**

- `log_wavelength_data:` Logarithmically spaced wavelength data.
- `bin_size:` The average difference in logarithmic wavelength, used for binning.
- `specids_list:` A list of the specids for the galaxies/QSOs processed.

4.2 Dereddening the Flux (`deredden_flux` function)

This function corrects for the effects of dust reddening in the observed spectra. This uses the extinction ccm89 model^[5]. Here A_V : extinction in magnitudes at characteristic V band wavelength and R_V is the ratio of total to selective extinction, $A_V / E(B-V)$. The mean R_V dependent extinction law is parameterized as

$$\langle A(\lambda)/A_V \rangle = a(x) + \frac{b(x)}{R_V}$$

where the coefficients $a(x)$ and $b(x)$ are functions of wavelength. At a wavelength of approximately 5494.5 angstroms (a characteristic wavelength for the V band), $a(x) = 1$ and $b(x) = 0$, so that $A(\lambda = 5494.5\text{\AA}) = A_V$. This function returns

$$A(\lambda) = A_V \left(a(x) + \frac{b(x)}{R_V} \right)$$

- **Arguments:**

- **reddened_data:** The spectral data affected by dust reddening.
- **limit:** The number of galaxies or QSOs for which dereddening is to be applied.

- **ebv_array**: An array containing the E(B-V) values, which quantify the dust extinction for each galaxy or QSO.
- **Process:**
 - **Extinction Curve Creation**: Using the 'extinction' package's ccm89 model, the function generates an extinction curve based on the wavelength and E(B-V) values.
 - **Dereddening Calculation**: The function applies the extinction curve to the reddened data, effectively removing the reddening effect.
- **Returns:**
 - **dereddened_data**: The flux data corrected for dust extinction.

4.3 Wavelength Range Normalization (**same_range** function)

This function standardizes the spectral data to a specified wavelength range, ensuring that all data sets are comparable. This needs to necessarily lie in the optical range.

- **Arguments:**
 - **min_wavelength**: The minimum wavelength value for the desired range.
 - **max_wavelength**: The maximum wavelength value for the desired range.
 - **data**: The spectral data to be cropped.
- **Process:**
 - **Data Cropping**: The function crops the spectral data to fit within the specified wavelength range.
- **Returns:**
 - **normalized_data**: Spectral data restricted to the specified wavelength range.

4.4 Binning (**bin_make** and **binning** functions)

These functions manage the creation of wavelength bins and the assignment of data points to these bins.

4.4.1 **bin_make** function

- **Arguments:**
 - **bin_size**: The bin size calculated during data acquisition.
 - **min_wavelength, max_wavelength**: The minimum and maximum wavelength values for the range.

- **Process:**
 - **Bin Creation:** The function calculates the bin edges and centers based on the given bin size and wavelength range.
- **Returns:**
 - **bin_edges:** The edges of the bins.
 - **bin_centers:** The centers of the bins.

4.4.2 binning function

- **Arguments:**
 - **bin_edges:** The edges of the bins created by the `bin_make` function.
 - **data:** The dereddened spectral data to be binned.
- **Process:**
 - **Data Assignment:** The function assigns data points to the appropriate bins based on their wavelength.
- **Returns:**
 - **binned_data:** The spectral data organized into bins.

4.5 Sigma Clipping (`sigma_clipping` function)

This function improves the SNR by removing anomalies from the data using a statistical approach. The understanding for the following is taken from GNU^[6]. σ -clipping is one commonly used way to remove/clip the effect of such very strong outliers in measures. σ -clipping is defined as the very simple iteration below. In each iteration, the number of input values used might decrease. The outliers will be removed through this iteration.

1. Calculate the standard deviation (σ) and median (m) of a distribution. The median is used because the mean is too significantly affected by the presence of outliers.
2. Remove all points that are smaller or larger than $m \pm \alpha\sigma$
3. Go back to step 1, unless the exit criteria is reached. The common type of exit criteria (to stop the σ -clipping iteration) is using the tolerance level. The tolerance level is defined by:

$$\frac{\sigma_{\text{old}} - \sigma_{\text{new}}}{\sigma_{\text{new}}}$$

- **Arguments:**
 - **data:** The binned spectral data.

- **n**: The number of standard deviations used to define the clipping threshold.
- **tolerance**: The acceptable level of variance for clipping.
- **Process**:
 - **Recursive Clipping**: The function iteratively removes data points that deviate beyond the defined threshold until the tolerance level is achieved.
- **Returns**:
 - **clipped_data**: The spectral data after sigma clipping.

4.6 Inverse Variance Calculation (**inv_var** function)

This function calculates the inverse variance for the binned data, which serves as an error measure.

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

- **Arguments**:
 - **data**: The clipped spectral data.
- **Process**:
 - **Inverse Variance Calculation**: The function calculates the inverse variance for each bin.
- **Returns**:
 - **inverse_variance_data**: The inverse variance for the binned data.

4.7 Weighted Mean Calculation (**weigh_mean** function)

This function calculates the weighted mean of the binned data, using the inverse variance to account for the reliability of each data point.

Given:

- f_i = the i -th measurement
- σ_i^2 = variance of the i -th measurement
- $w_i = \frac{1}{\sigma_i^2}$ = weight associated with the i -th measurement

The weighted mean $\bar{f}_{\text{weighted}}$ is calculated as:

$$\bar{f}_{\text{weighted}} = \frac{\sum_{i=1}^n w_i f_i}{\sum_{i=1}^n w_i}$$

where n is the number of measurements.

- **Arguments:**
 - **data:** The binned spectral data.
 - **inverse_variance:** The inverse variance calculated for the data.
- **Process:**
 - **Weighted Mean Calculation:** The function computes the weighted mean, giving more weight to data points with lower variance.
- **Returns:**
 - **weighted_mean_data:** The weighted mean of the spectral data.

5. Results

Using the code described two samples are created. The first sample as also given in the GitHub link for the code is of first 100 QSOs taken from SDSS-DR16 sorted on the basis of redshift (redshift: [0.6,1.1], declination: [0,10], right ascension: [0,10]). The second sample compiles the result for first 100 QSOs from BOSS-DR16 sorted on the basis of redshift (redshift: [0.6,1.1], declination: no filter, right ascension: no filter). For each sample two plots are given: one using the simple mean ([Figure 1.1](#) & [Figure 1.2](#)) and others using the weighted mean ([Figure 2.1](#) & [Figure 2.2](#)). The weighted mean plot did not match expected emission line spectra for both, giving an abrupt graph in the second sample, possibly due to bias or incorrect weights, while the simple mean plot showed recognizable emission lines, suggesting it might be more reliable for this data. In simple mean plot the spectral emission lines are identifiable.

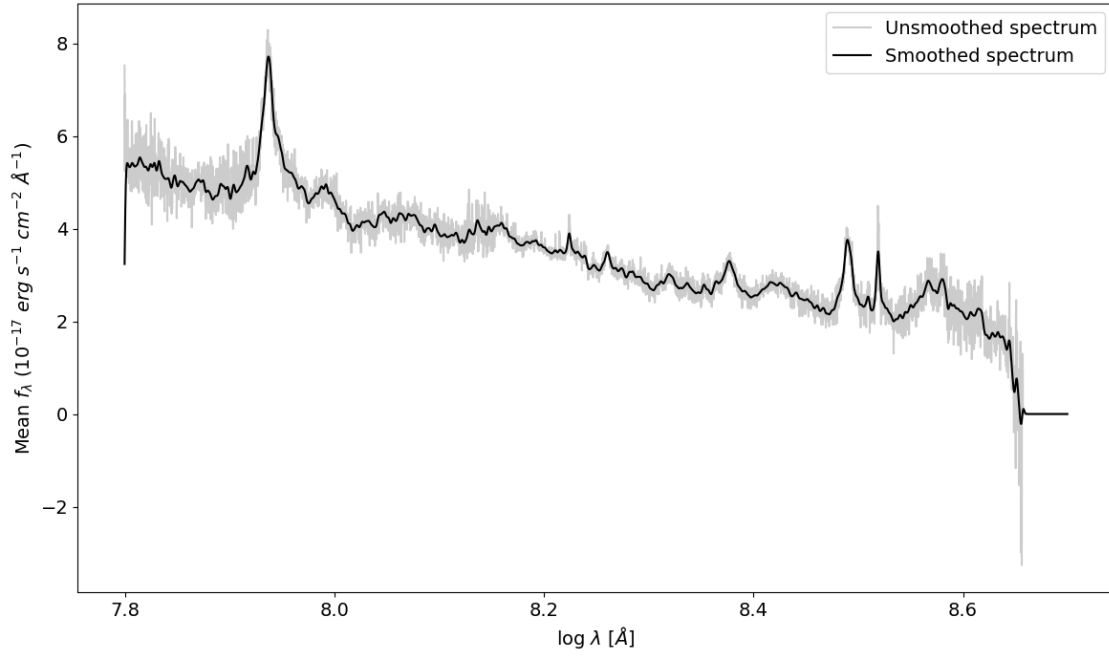


Figure 1.1: Plot using the weighted mean.

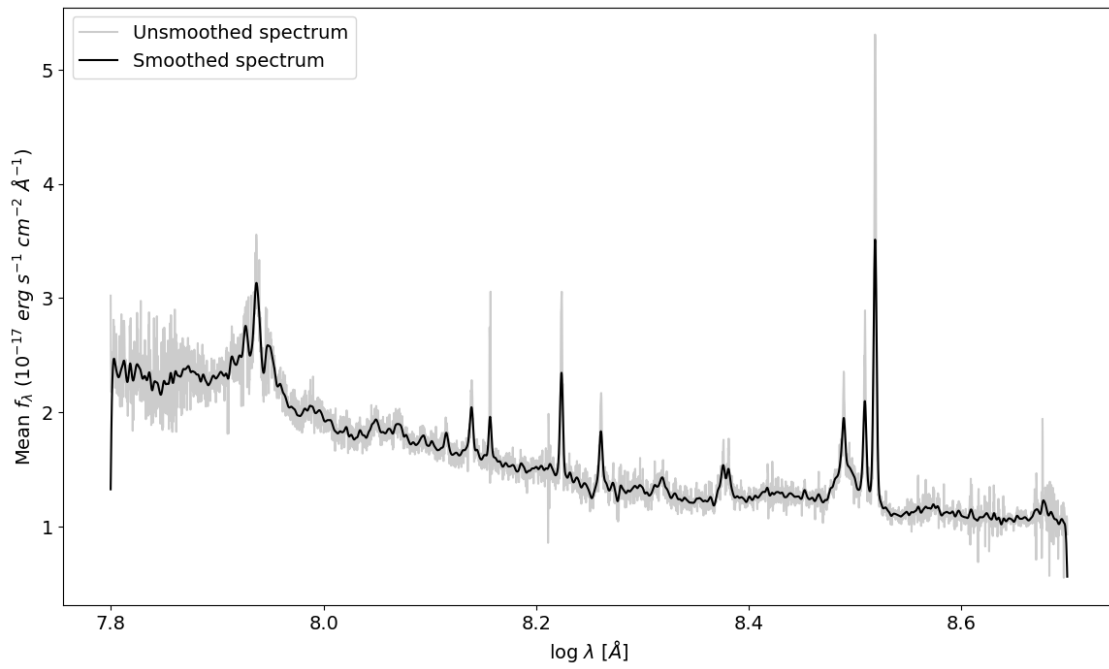


Figure 1.2: Plot using the weighted mean.

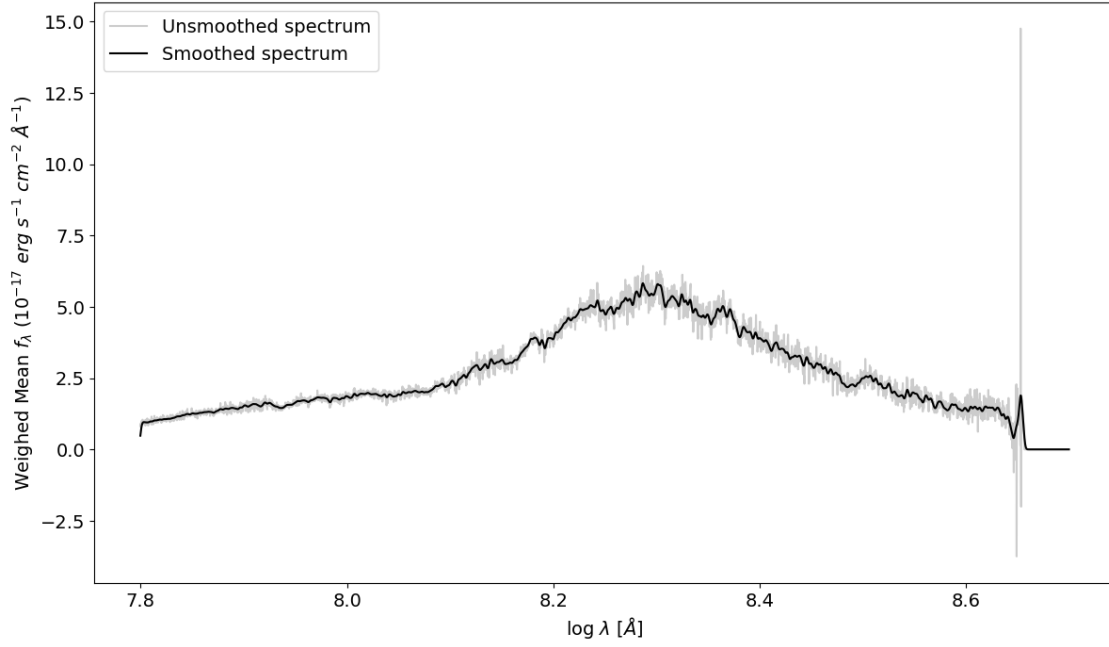


Figure 2.1: Plot using the weighted mean.

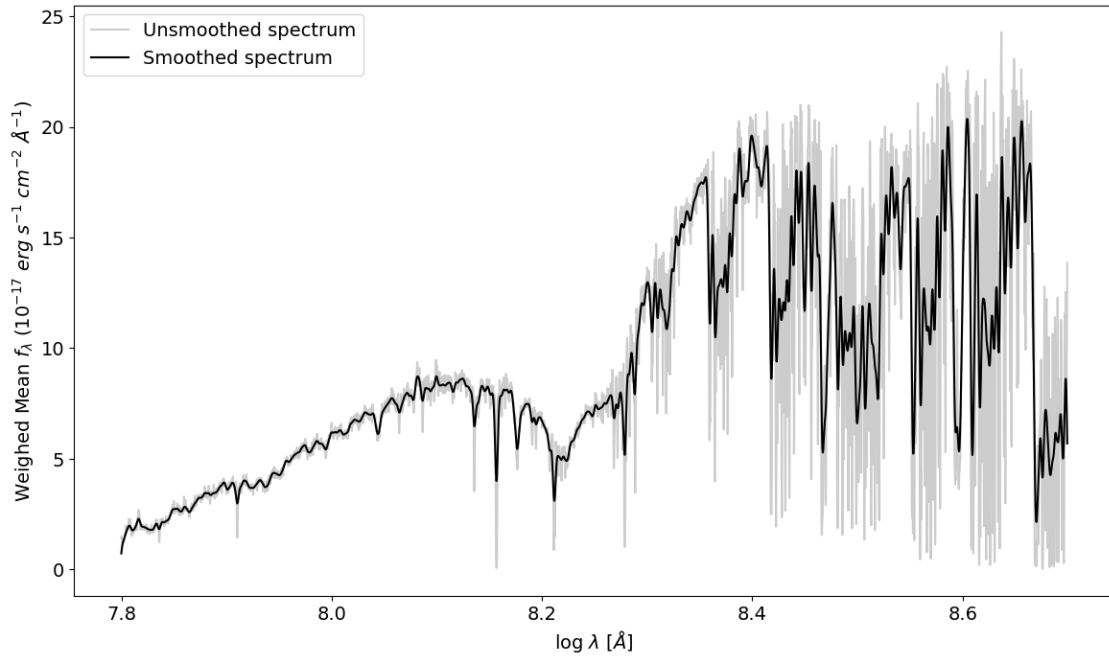


Figure 2.2: Plot using the weighted mean.

6. Skills Gained

- **Conceptual Skills:** Deepened understanding of spectral analysis, AGN properties, and the significance of stacking spectra in astrophysics. Attended special lectures organized for group members for increased understanding of complex topics like Kick Velocity of black hole mergers.
- **Technical Skills:** Enhanced proficiency in Python, Jupyter notebooks, and the application of astronomical data analysis packages. Got to deal with simulations of black hole mergers to find their kick velocities. Gained experience with large data set and its compilation through packages like SPARCL.
- **Collaborative Skills:** Experience in integrating data from multiple surveys and collaborating with team members on code development through GitLab. A brief understanding on how to write a report was also developed.

7. Conclusion

This project successfully developed a code capable of stacking spectra for multiple galaxies and QSOs simultaneously. However, the results suggest that the weighted mean may not always produce accurate results, possibly due to biases or incorrect weight assignments. The simple mean, in contrast, produced plots that matched expected emission line spectra more closely, indicating its potential reliability for this type of analysis.

8. References

- 1 [Jed McKinney et al \(2023\)](#), "Broad Emission Lines in Optical Spectra of Hot, Dust-obscured Galaxies Can Contribute Significantly to JWST/NIRCam Photometry"
- 2 [sparclclient GitHub repository](#)
- 3 [Dawson et al. 2016](#), "The SDSS-IV Extended Baryon Oscillation Spectroscopic Survey: Overview and Early Data"
- 4 [DESI Collaboration et al. \(2023b\)](#), "The Early Data Release of the Dark Energy Spectroscopic Instrument"
- 5 Kyle Barbary, [Extinction GitHub repository](#) "extinction ccm89 model"
- 6 [GNU Astronomy Utilities](#), "Sigma Clipping"